

# BÓBR AIR

A desk climate station you build  
and program yourself.



Sto lat, Szymon!

OCTOBER 2026

# Cześć Szymon!

We've spent a lot of hours in front of screens together. So this time I made you one you build yourself.

In this box is a small desk station. It measures the temperature, the humidity, the air pressure and the air in your room. Building it takes about an hour, with four solder joints and twelve screws. The software is up to you: write your own with Claude, or install the finished one.

When the air gets stale, the bóbr tells you. Then open a window, or even better, go for one of your long walks.

## Sto lat!

### Remus

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#### BEFORE YOU START

- The soldering iron gets 350 °C hot. Always put it back in its stand.
- Open a window while you solder. Your new sensor will notice the smoke too.
- Work on a soft towel, so the screen doesn't get scratched.

**1 h**

to build

**4**

solder joints

**12**

screws

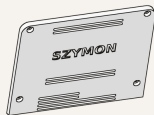
**2**

ways to  
program it

# In the box



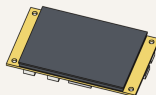
- 1**  
**Front case**  
with a bóbr on it



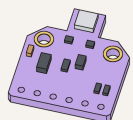
- 2**  
**Back lid**  
with your name



- 3**  
**Base**  
the case stands on it



- 4**  
**Screen board**  
ESP32 with a 2.8 inch touch screen



- 5**  
**Sensor**  
Bosch BME680,  
measures the air



- 6**  
**Sensor cable**  
white plug, 4 wires,  
plus spare Dupont  
wires



- 7**  
**Śrubki box**  
M2 screws for each  
part and rubber  
feet, all with spares



- 8**  
**USB cable**  
with an angled plug



- 9**  
**Soldering set**  
iron, stand and  
solder



- 10**  
**Screwdriver**  
small, for the M2  
screws

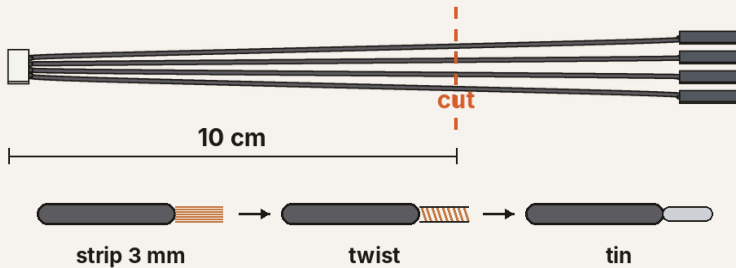
## YOU ALSO NEED

A laptop (for step 7), scissors or side cutters, and a USB charger.  
For option 1 also a paid Claude plan.

*And a book, to read while the sensor burns in.*

## 1 Cut and strip the cable

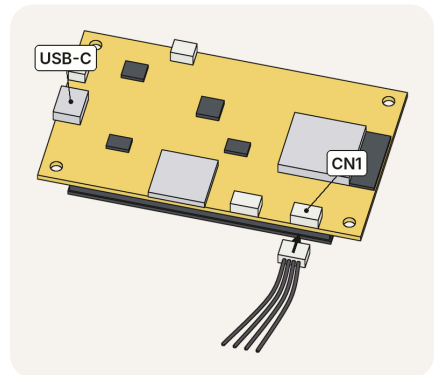
Cut the cable 10 cm from the white plug. You don't need the black plugs. Strip 3 mm from each wire, twist the strands and tin them.



## 2 Find out which wire is which

Plug the white plug into **CN1** on the back of the screen board and leave it there. The pin names are printed next to CN1. Follow each wire and write its colour here.

| CN1  | COLOUR | SENSOR |
|------|--------|--------|
| 3.3V |        | VCC    |
| GND  |        | GND    |
| I022 |        | SCL    |
| I027 |        | SDA    |

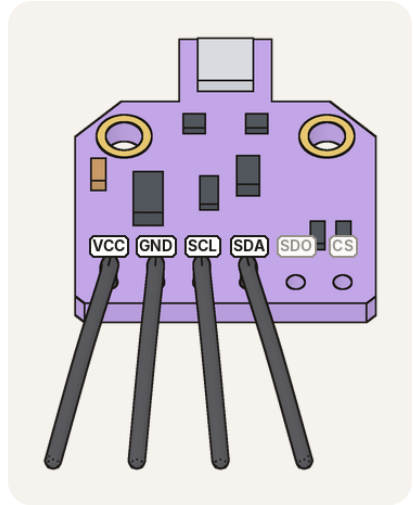


### 3 Solder the sensor

Push each wire through its hole from the side with the parts. Solder it on the flat side, then cut off what sticks out.

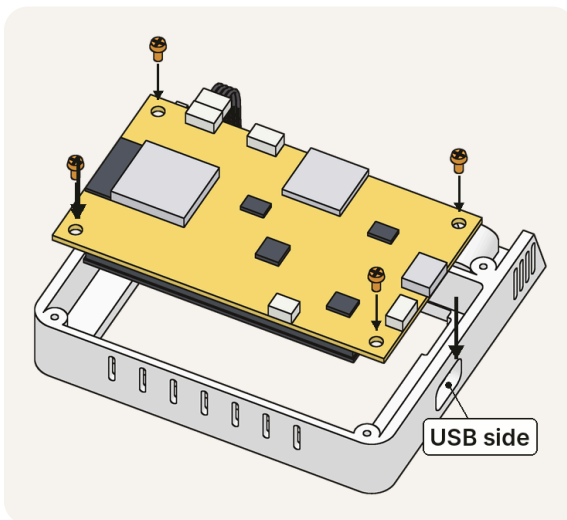
**SDO** and **CS** stay empty.

Tip: 350 °C with the chisel tip. Heat the pad and the wire for 1 to 2 seconds, then add the solder. A good joint is shiny and looks like a small volcano.



### 4 Screen board into the front

Lay the front case face down on a soft towel. Put the board in with the screen down and the USB port at the side opening. The white plug sits in the small pocket under the board.

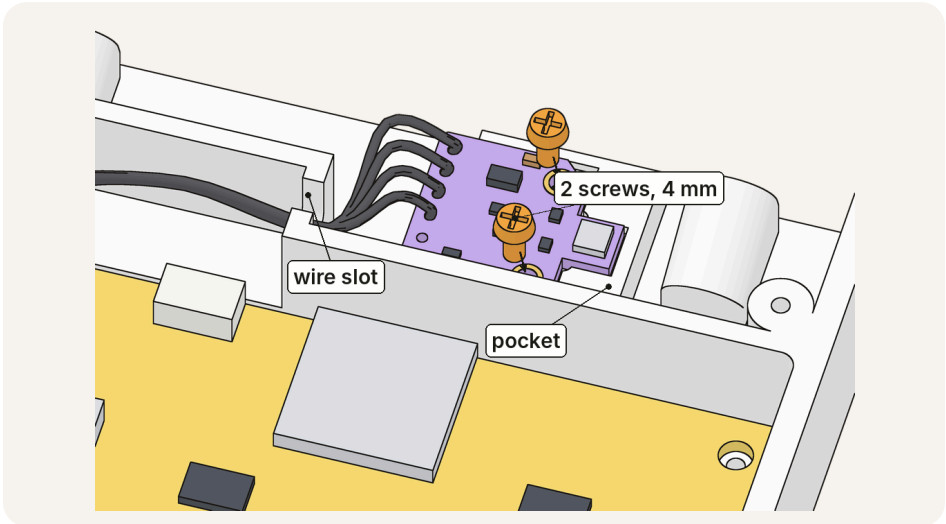


Fix it with the 4 screws from **SCREEN** (4 mm) and the small screwdriver.

Snug, not tight. The screws cut their own thread into the plastic.

## 5 Sensor into its own room

Lay the sensor into the T-shaped pocket in the lower room: chip up, pins towards the wire slot. It only fits one way. Fix it with the 2 screws from **SENSOR** (4 mm), snug, not tight. Lay the wires into the slot.

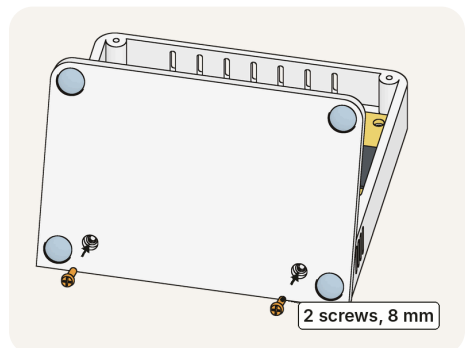


A small piece of paper tape over the slot keeps warm air from the board away from the sensor. No Kapton tape and no silicone near the sensor: they can damage it for good.

## 6 Base and feet

Stand the case on the base, front edges flush. Fix it from below with the 2 screws from **BASE** (8 mm). The lid stays off for now.

Then press the 4 rubber feet into the 4 round pockets under the base. They keep it from sliding when you tap the screen.



## 7 Bring it to life

The case stands on its base, the lid is still off. Plug a USB cable that carries data into the side and into a laptop. Pick one way:

### OPTION 1 • RECOMMENDED

#### Write your own software with Claude

Claude knows this board. You say what you want, it writes the code and flashes it. Set it up as on **page 11**, then say: **“Flash the starter and check my hardware.”**

### OPTION 2 • NO CODE

#### Install the finished BÓBR AIR

Scan the code, open the page on the laptop in Chrome or Edge and click the button for your board. About 2 minutes.



scan me

#### Check before you close the lid.

The starter says *Sensor OK* and shows your first numbers. BÓBR AIR shows the numbers, and air quality says *Starting*.

*Not found or No sensor?* Check the four joints and your colour table now, while the lid is off. Then unplug USB.

Hej Szymon!

Starter kit: hardware check

Screen **OK** if you can read this

Sensor **OK** BME680 at 0x77

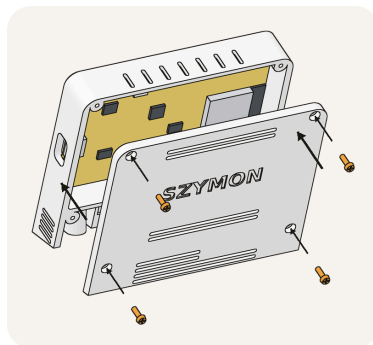
Touch **OK**

22.6°C 47% 1013 hPa

Your turn. Tell Claude what to build.

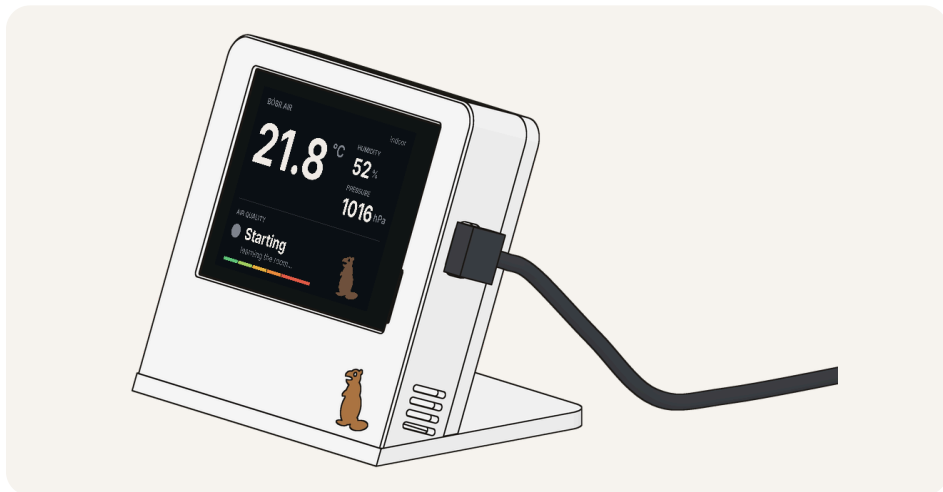
## 8 Close the lid

Unplug USB. Put the lid on with your name outside. Fix it with the 4 screws from **LID** (6 mm). Their heads sink into the lid.



## 9 Switch it on

Plug the angled cable into the side, with the cable pointing back, and into a USB charger.



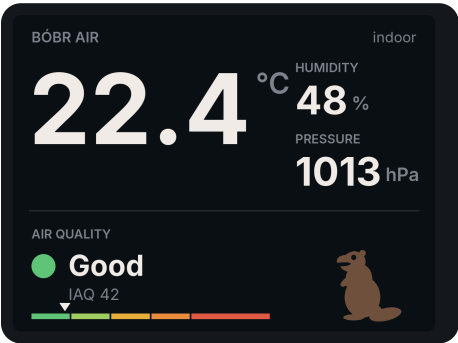
Board with two USB ports: use a USB-A to USB-C cable and a USB-A charger. USB-C to USB-C doesn't work on this board.

**Done. Put it where you sit, and give it two days to learn your room.**

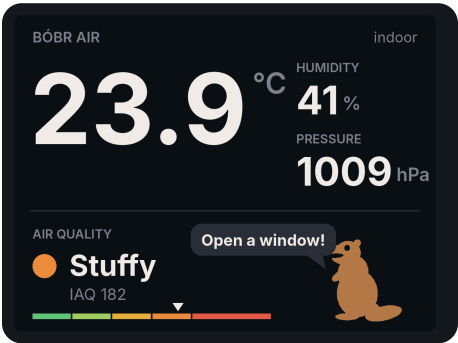
OPTION 2

# The BÓBR AIR screen

Temperature, humidity and pressure are right from the start. Air quality needs time: for the first two days the value moves a lot while the sensor burns in and learns your room. Until it is sure, it says *still learning*.



All good.



Stuffy: the bóbr wants fresh air.

## AIR QUALITY (IAQ)

- 0 to 50 Good
- 51 to 100 Fine
- 101 to 150 Stale
- 151 to 200 Stuffy. *Open a window!*
- over 200 Bad. *Time for a walk!*



Bad: close the laptop, go outside.

It saves what it has learned every 6 hours, so unplugging is fine. No WiFi, no app, no account.

# If something is wrong

## The Install button does nothing, or there is no port

Use Chrome or Edge on a laptop and try another USB cable: many only charge. On Windows the board may need the CH340 USB driver.

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## Claude can't upload to the board

Hold BOOT on the board while it says *Connecting*.

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## A board with two USB ports stays dark on USB-C

Its USB-C port only works with a USB-A to USB-C cable. The micro-USB port works with any data cable.

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## The screen is white, or the colours look strange

It has the software for the other board type. BÓBR AIR: install with the other button. Your own: tell Claude how many USB ports your board has.

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## The picture is upside down

BÓBR AIR: hold a finger on it for 3 seconds. Your own: ask Claude.

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## It says No sensor or Not found

Check the four joints and the wire colours in your table on page 4.

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## Your own software shows nothing

Install BÓBR AIR (page 7, option 2). If that works, the hardware is fine and the problem is in the code.

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## The temperature looks a bit high

Give it 30 minutes after switching on. Keep it out of the sun and away from the warm air of a laptop.

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## Still stuck?

Everything is also on [remustoma.github.io/bobr-air](https://remustoma.github.io/bobr-air). Or call Remus.

## OPTION 1

# Write your own software

The hardware is done, the software is yours. Claude Code knows this board: the pins, the screen, the sensor and how to flash it. You say what you want. Claude writes the code, flashes it and reads the board's log.

## SET IT UP

- a Install Claude Code. It needs a paid Claude plan.

```
curl -fsSL https://claude.ai/install.sh | bash
```

Windows, in PowerShell: `irm https://claude.ai/install.ps1 | iex`

- b Get the starter kit and start Claude in it:

```
git clone https://github.com/remustoma/bobr-air
cd bobr-air
claude
```

- c Connect the board and say: **“Flash the starter and check my hardware.”** Claude asks which board you have and does the rest.

## THEN ASK FOR ANYTHING

“ Count my weeks: I was born on [date]. Show them out of 5,218.

“ Make the bóbr speak Polish.

“ Show the outdoor temperature next to the indoor one.

“ Dim the screen at night with the light sensor.

- Claude can't see the screen. Tell it what you see, or paste a photo.
- When something works, say **“commit this”**. Then you can always go back.
- Broke it? Option 2 on page 7 puts BÓBR AIR back on the board.
- Other AI tools work too. Their notes are in `CLAUDE.md` in the kit.



**4,000 weeks is the  
average. Sto lat is 5,218.  
Spend some of them  
outside.**

**BÓBR AIR • KIT 01 / 01**

**Designed by Remus**

**Built and coded by Szymon**

**October 2026**

**[remustoma.github.io/bobr-air](https://remustoma.github.io/bobr-air)**

